RSA

R → Rivest
S → Shamir
A → Adleman

RSA Steps:

① Randomly select 2 large prime numbers "p" & "q" such that $p \neq q$

② Calculate "n"

$n = p * q$

③ ϕ = "phi" $= (p-1)(q-1)$

$n = 6$

~~2~~ $2 * 3$

$p \quad q$



(4) Select value "e"

$1 < e < phi$ & some additional conditions

"e" ⇒ encryption exponent

(5) Calculate "d"

$(d)(e) \bmod phi = 1$

"e" ⇒ decryption exponent

The Public Key: (n, e)

The Private Key: (n, d)

Encryption:

$$c = m^e \mod n$$

Decryption:

$$m = c^d \mod n$$

Simple example of RSA

1. Select 2 prime numbers $p = 5$ and $q = 3$
2. $n = p * q = 5 * 3 = 15$
3. $phi = (p-1)(q-1) = (5-1)(3-1) = 4 \times 2 = 8$
4. e such that: $1 < e < phi$ and other conditions

$e = 3$

5. $d = ?$ $\Longrightarrow$ $\boxed{(d)(e) \bmod phi = 1}$

$(d)(3) \bmod 8 = 1$

$\boxed{d = 3}$

6. Pub: (n, e)
   Priv: (n, d)

Pub ⇒ $(n, \epsilon) = (15, 3)$

Pair ⇒ $(n, d) = (15, 3)$

$m = 7$

Encryption:

$$c = m^e \bmod n$$

$$c = 7^3 \bmod 15$$

$$\boxed{c = 13}$$

Decryption:

$$m = c^d \bmod n$$

$$m = 13^3 \bmod 15$$

$$m = 7$$

Example 2 RSA

① $p = 61$ & $q = 53$

② $n = p * q = 61 * 53 = 3233$

③ $phi = (p-1)(q-1) = (60) * (52) = 3120$

④ $e \hookrightarrow 17$ ✓
↳ 4000

$e = 17$

⑤ $d = 2753$

$(d)(e) \mod phi = 1$

$(2753)(17) \mod 3120 = 1$

$m = 65$

Encryption

$$c = m^e \bmod n$$

$$c = 65^{17} \bmod 3233$$

$$\boxed{c = 2790}$$

Decryption:

$$m = c^d \bmod n$$

$$m = 2790^{2753} \bmod 3233$$

$$m = 65$$